

Child Labor and Sibling Schooling: Evidence from 1920 America

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Abstract

If an older child's earnings relax the family budget constraint, access to work may allow younger siblings to remain in school. We test this mechanism using state minimum-working-age laws and the full-count 1920 US Census. For each younger child, we measure work and legal exposure only among older siblings and compare children of the same age with identical older-sibling age structures across states with different laws. Legal exposure strongly increased older-sibling work. The estimated effect of that work on younger children's attendance is 1.7 percentage points, with a state-clustered 95 percent confidence interval from -1.9 to 5.3 points. We also find no positive attendance effect among families that appeared economically vulnerable before 1920. These results differ from earlier household-level evidence. We show that the existing evidence comes mainly from families with only one child aged 10–16, where it cannot represent a transfer between siblings. Separating the child who works from the child who might benefit alters the conclusion: there is little support for the view that older children's employment financed younger siblings' schooling.

Keywords: child labor, school attendance, siblings, intrahousehold allocation, historical census

JEL codes: J13, J22, I21, N31

1 Introduction

Child labor may reduce one child’s schooling while helping finance another child’s schooling. The first effect is well established in the historical literature, while the second is a central implication of economic models of child labor and is consistent with historical accounts of the family economy (Basu and Van, 1998; Baland and Robinson, 2000; Parsons and Goldin, 1989). In households that have little access to credit or insurance, preventing an older child from working may reduce the resources available to support the education of siblings. Modern experimental evidence is consistent with this idea, and has shown that the income received by one household member can alter other members’ work and education choices (Rhodes et al., 2026). Here we explore the specific question of whether an older child’s earnings increase the amount of schooling that younger siblings obtain. Identifying this effect is difficult, however, because the conditions that influence the likelihood that a child works also influence the likelihood that their siblings go to school.

The answer to this question is critical for understanding the incidence of child-labor policy. Existing evidence appears to support the family-insurance mechanism. Most prominently, Manacorda (2006) instruments the share of children in a family doing market work with their average exposure to child labor law thresholds and finds that greater family work increased sibling school attendance. He finds that for a family with two children aged 10–16, moving one child into work raised sibling school attendance by roughly 6.5 percentage points. However, we show that the household work measure and instrument that are used are not appropriate. Specifically, they both include the siblings whose schooling is potentially affected. Moreover, families with only one child aged 10–16 are also included in both measures and this variation contributes to the identified effect, even though it should not.

We revisit the sibling-schooling question using state minimum-working-age laws and the full-count 1920 US Census. The full count allows us to replace the household average with the specific sibling relationship that is exposed to the law. For each younger child, we measure work and legal exposure only among older siblings aged 10–16. Our preferred specification then compares younger children of the same age who have the same number of older siblings who are exactly the same ages. As a result, their sibling age structure is identical, but differences in state law generate different amounts of legally induced older-sibling work.

Our first result is that the laws strongly predict older-sibling work, consistent with a long historical literature (Moehling, 1999; Lleras-Muney, 2002; Clay et al., 2021). However, the preferred estimates show no detectable increase in schooling among younger siblings. In a sample of 1.76 million children, our preferred estimate is 1.7 percentage points, with a 95 percent confidence interval that excludes the Manacorda (2006) estimate of 6.5 percentage points. We further test whether the result depends on the specifics of the child labor policy. Allowing more flexible state-specific sibling-age profiles leaves much less identifying variation and cannot rule out a substantial benefit.

We also explore why the household- and individual-level estimates differ. We first recover the household coefficient estimated in Manacorda (2006) from the original one-percent sample and show that we can obtain the same coefficient in the full count. The difference is therefore not due to the

details of the data construction. Instead, families with one child, who should be excluded, receive 27 percent of the two-stage least-squares weight but generate 89 percent of the positive school attendance estimate. We show that excluding families with only one child eliminates the finding of a spillover across siblings. We then test whether this average effect conceals a spillover benefit among specific types of families, in particular those that are most likely to depend on child earnings. We link families to the 1910 Census and measure economic vulnerability before the 1920 work and schooling decisions. The sibling estimate remains near zero among families with low paternal occupational standing and among families in which at least one parent was illiterate. Again, legal exposure strongly predicts older-sibling work within these groups, but the additional work does not increase sibling school attendance.

The paper connects research on child-labor regulation to evidence on household resources and sibling spillovers. Work on compulsory-schooling and child-labor laws asks how legal restrictions affected the education of the child directly covered by the law (Moehling, 1999; Lleras-Muney, 2002; Clay et al., 2021). Evidence on household income provides a reason to expect effects on other children as well. In South Africa, Edmonds (2006) finds that pension eligibility for elderly men increased children’s school attendance and reduced their hours of work. More broadly, the form and frequency of transfers can shape household spending (Bonomo et al., 2026). Research on sibling spillovers also shows that shocks to one child can affect the outcomes of other children (Black et al., 2021). These findings make sibling financing plausible, but they do not establish that additional employment by an older child increases a younger sibling’s attendance.

We make the cross-child comparison explicit. We then reproduce the household result, show which family structures identify it, and finally reconstruct the treatment to target the sibling whose work should generate the spillover benefits. These steps separate the response of the child covered by the law from effects operating through other children.

The remainder of the paper is organized as follows. Section 2 describes the historical setting, and Section 3 introduces the data and samples. Section 4 develops the empirical design, and Section 5 presents the older-to-younger estimates. Section 6 explains why the household-average estimate differs, and Section 7 examines economically vulnerable families. Section 8 concludes.

2 Child Labor and Schooling in 1920

By 1920, two large and related changes were reshaping childhood in the United States. Child labor had declined substantially from its late nineteenth-century peak, while the high school movement was expanding access to secondary education. Work nevertheless remained common among adolescents, particularly in farm households, and continued schooling beyond elementary school was far from automatic. Children in their early teens could attend school, work, combine the two, or do neither. How to use an adolescent’s time was therefore an important household decision (Goldin and Katz, 2008, 2011).

This setting meant that children typically contributed labor and earnings to households that

had limited access to formal credit and little public insurance against lost income. Importantly, historical accounts treat child work as part of a family allocation problem rather than as an isolated choice by the child (Parsons and Goldin, 1989). In a simple model, a binding household resource constraint can make child labor privately necessary even when schooling has a high return (Basu and Van, 1998; Baland and Robinson, 2000; Edmonds, 2007). If an older child’s earnings relax that constraint, the family can use some of the additional resources to keep a younger child in school. This is the specific household response that we test.

The federal government had also begun to regulate child labor by 1920, though we do not exploit national-level variation. Specifically, Title XII of the Revenue Act of 1918 taxed firms that employed children below specified ages,¹ though it excluded the bulk of agricultural and unpaid family work. State work-permit and compulsory-attendance laws therefore continued to determine the age at which market employment became legally available and this is the cross-state variation that we exploit. Our outcomes include household-farm work and mutually exclusive combinations of work and school.

There was substantial heterogeneity across states in child labor laws. Specifically, school-entry ages, compulsory-attendance requirements, educational exemptions, and work-permit rules often overlapped, and enforcement varied across states. This institutional complexity helps explain why historical studies disagree about how much legislation contributed to the decline of child labor. Moehling (1999) finds a limited role for early child-labor laws, whereas Lleras-Muney (2002) and Clay et al. (2021) find that compulsory-attendance and child-labor laws increased education. Work on later child-labor regulation also shows that households can respond along margins not directly targeted by the law (Bharadwaj et al., 2020). These studies primarily establish effects on the children directly exposed to regulation.

Following Manacorda (2006), we use the state minimum age for a work permit, a dataset originally assembled by Lleras-Muney (2002). Most states fall into two groups. Seven states allowed work permits at age 12, while 32 set the minimum at age 14. Four states used age 15, one used age 13, and four had a lower or unrecorded minimum. New Mexico has no recorded minimum and is coded as zero, Louisiana and Wyoming are coded as age 7, and Florida as age 8. These differences generate the first-stage variation in the empirical design. Appendix Table 8 reports the work-age, compulsory-schooling, educational-exemption, work-permit, and continuation-school provisions for every state.

For a child of age a in state s , legal exposure is

$$E_{is} = \max\{a_i - L_s + 1, 0\}, \quad (1)$$

where L_s is the minimum work age in state s . Exposure is zero below the threshold and increases

¹Approved in February 1919, Title XII imposed a tax equal to 10 percent of net profits on mines and quarries employing children under age 16 and on factories, mills, workshops, and canneries employing children under age 14 or employing children aged 14–16 beyond specified hours (United States Congress, 1919). The tax was in force during the 1920 census reference period and was struck down in 1922 in *Bailey v. Drexel Furniture Co.* (U.S. Supreme Court, 1922).

by one in each year thereafter. The measure therefore captures not only whether a child is legally eligible to work, but also how long work has been available. Since age is strongly related to both work and schooling, every specification controls flexibly for the focal child’s age and for the child’s own legal exposure within each legal regime.

The family-financing interpretation yields a simple sequence of predictions. State law should first increase work among older children. That work should then increase school attendance among their younger siblings, with the largest response in families least able to replace the lost earnings. The empirical analysis explores these margins in turn.

3 Data and Samples

3.1 The 1920 Census

Our main sample comes from the complete 1920 Census distributed by IPUMS ([Ruggles et al., 2025](#)). The existing household evidence in [Manacorda \(2006\)](#) instead uses the one-percent IPUMS sample. We use the full count not only to increase precision, but also to make a different empirical comparison possible. The additional observations allow us to compare younger children with identical older-sibling age rosters and to measure work and legal exposure only among those older siblings. These comparisons are too sparse to estimate precisely in the one-percent sample.

To maintain comparability with the existing household evidence, we adopt the family restrictions applied in [Manacorda \(2006\)](#). The sample consists of children aged 10–16 who lived with both parents in family units containing only the parents and children aged 0–16. Mothers were required to be between 25 and 41 years old and fathers between 25 and 66. These are restrictions on the census family unit, not on other family units sharing the household. Appendix Section D.2 relaxes the parental-age and family-composition restrictions while retaining two identifiable co-resident parents.

Market work is defined using the harmonized 1950 occupation code and excludes unpaid work on the household farm. We also report a broader measure of any recorded work that includes the farm categories. The census asked whether a person had attended school, college, or any educational institution at any time between September 1, 1919 and January 1, 1920. Our outcome therefore measures recent attendance, not completed schooling or daily enrollment. Work is recorded from the reported occupation, so work and attendance can overlap. Children are classified as idle when neither work nor attendance is recorded.

The resulting sample contains 4,428,417 children in 2,652,969 families. School attendance was 88.7 percent, 8.7 percent of children reported some work, and 7.1 percent were neither working nor in school. The average market-work share among children aged 10–16 was 8.5 percent. Appendix Table 6 reports the distribution of the sample across family sizes.

We also exploit the raw 1920 census extract deposited with [Manacorda \(2006\)](#) to reconstruct the household benchmark. Applying the published restrictions and variable definitions produces 44,451 children in 26,645 families, compared with 44,402 children in 26,612 families reported in the article.

The difference of 49 children has no effect on the estimates. A deterministic one-percent draw from our full-count family sample contains 44,448 children in 26,635 families, providing a separate check on the construction. The archived estimation program for the paper appears to implement an earlier specification rather than the one reported in the published article. It uses different sample restrictions, a different definition of work, and a single binary instrument rather than the seven instruments reported in the article. We therefore reconstruct the published specification directly from the deposited raw data. Appendix A and Table 4 document the reconstruction and the differences between the archived program, the published counts, and the full-count implementation.

3.2 Linked 1910 backgrounds

For the final analysis, we add characteristics measured ten years before the 1920 outcomes. Census Tree links connect children and their parents to the 1910 Census (Buckles et al., 2025). When several children identify the same parents, we require that they link to a common 1910 family. The resulting sample contains 1,217,941 children in 846,145 families.

These linked families are not representative of the full 1920 sample. They are more likely to have a farm background and to have children in school, and they are less likely to have an illiterate father. We therefore use the full count for the main estimates and the linked sample only to compare families with different preexisting characteristics. As a robustness check, we re-weight linked families to match the observable composition of the eligible full-count 1920 sample. The weights balance state, sampled-child age structure, the number of younger children, sex composition, urban and farm status, race, parental ages, literacy, immigrant status, and housing tenure. This exercise addresses selection on these observed characteristics, although not on unobserved determinants of census linkage. Appendix Section E describes the procedure and reports the reweighted estimates (which are qualitatively similar to the unweighted estimates).

3.3 Linked 1940 outcomes

We also follow the younger children to the 1940 Census to examine completed education, using the historical identification keys from the IPUMS Multigenerational Longitudinal Panel. We retain unique links with consistent sex and an age increase of 20 years, allowing two years of reporting error. Work and legal exposure are constructed from the complete 1920 sibling roster before selecting linked children. Because linkage is much more common for boys, the attainment analysis focuses on sons. Appendix Section D.3 reports completed years, grade 8 and grade 12 completion, linkage diagnostics, and estimates reweighted to the observable composition of eligible 1920 sons.

4 Empirical Design

Our main design follows the proposed flow of resources from older to younger children. Let i index a focal child in family f and state s , and let Y_{ifs} denote that child’s outcome. Let $j = 1, \dots, N_{if}^O$ index child i ’s older siblings aged 10–16, where N_{if}^O is the number of such siblings. The sample of

primary interest is restricted to children with $N_{if}^O \geq 1$. For each sibling j , W_{jf} equals one if the census records market work and zero otherwise. Legal exposure, E_{js} , is defined as in (1) above. We define the corresponding older-sibling averages as

$$\overline{W}_{if}^O = \frac{1}{N_{if}^O} \sum_{j=1}^{N_{if}^O} W_{jf}, \quad \overline{E}_{if}^O = \frac{1}{N_{if}^O} \sum_{j=1}^{N_{if}^O} E_{js}, \quad (2)$$

Thus, $\overline{W}_{if}^O$ is the share of older siblings who work, while $\overline{E}_{if}^O$ is their average number of years of legal exposure. The preferred estimating equation is

$$Y_{ifs} = \beta \overline{W}_{if}^O + X'_{ifs} \gamma + \lambda_{sa_i} + \mu_{R_{if}^O} + \varepsilon_{ifs}, \quad (3)$$

where R_{if}^O is the complete roster of older-sibling ages. The terms λ_{sa_i} and $\mu_{R_{if}^O}$ denote state-by-focal-age and exact-roster fixed effects, respectively. The excluded instruments are

$$Z_{ifr}^O = \overline{E}_{if}^O \mathbf{1}\{L_s = r\}, \quad r \in \{0, 7, 8, 12, 13, 14, 15\}. \quad (4)$$

Here $\mathbf{1}\{L_s = r\}$ indicates that the minimum work age in state s equals r . In a state with a minimum work age of 14, for example, only $Z_{if,14}^O$ is nonzero. Interacting exposure with the seven observed regimes allows the relationship between legal exposure and work to differ across the state-law thresholds. The first stage therefore asks whether the same set of older-sibling ages produced more work under a state law that gave those siblings more years of legal access to employment.

The preferred specification includes state-by-focal-age fixed effects and fixed effects for the complete roster of older-sibling ages. It therefore compares younger children of the same age whose older siblings are also exactly the same ages, but whose state laws generate different legal exposure. Identification requires that, absent differences in child-labor laws, the relationship between older-sibling ages and younger-child attendance would not vary across states in ways that are correlated with the legal working-age thresholds. We assess this assumption by looking at the school attendance patterns around each threshold and by allowing the relationship between older-sibling age and younger-child attendance to vary flexibly across states.

We control for the exact age profile of siblings in order to avoid the ordinary fact that a 16-year-old sibling is both more likely to work and be observed in a household at a later stage of family life relative to an 11-year-old sibling. The remaining controls include the child's sex and birth order, the number and ages of younger children, family composition, urban and farm status, parental characteristics, housing tenure, race, and family size. Standard errors are clustered by state, the level at which the laws vary. We also report a wild-cluster bootstrap, leave-one-state-out estimates, a state-law permutation diagnostic, and Anderson–Rubin tests.

Importantly, we interpret the instrumental-variables estimate as the broad effect of law-induced older-sibling employment rather than as the effect of household income alone. This is in part because an older child's employment may also alter caregiving within the household, parental labor supply,

or household expectations.

The coefficient is a weighted average of responses among families whose older-sibling work changed with legal exposure. We report each law-specific first stage and reduced form, its implicit weight, and estimates that omit one state law regime at a time. We also report separate estimates for the 39 states with the common age thresholds of 12 and 14, for which the comparison is easiest to see.

5 Older-Sibling Work and Younger Children’s Schooling

Table 1 reports the effect of older-sibling work on younger children. The preferred specification appears in the third row, which holds fixed the focal child’s age and the exact roster of older-sibling ages. Legal exposure strongly predicts older-sibling work, and the attendance point estimate is 0.017. The table bases inference on 48 state clusters and reports the outcome mean. Appendix Table 10 reports the law-specific first-stage coefficients and their state-clustered standard errors.

Table 1: Does Work by Older Siblings Affect Younger Children?

Specification	Any work	In school	Neither
State fixed effects	0.003 (0.009)	−0.025*** (0.008)	0.037*** (0.005)
State by focal-child age	0.008* (0.005)	−0.025*** (0.006)	0.027*** (0.004)
Exact older-sibling age roster	0.013 (0.016)	0.017 (0.018)	−0.008 (0.014)
Outcome means, preferred sample	0.057	0.927	0.057
Older-sibling market-work share		0.169	
Observations		1,756,540	
Families		1,242,863	

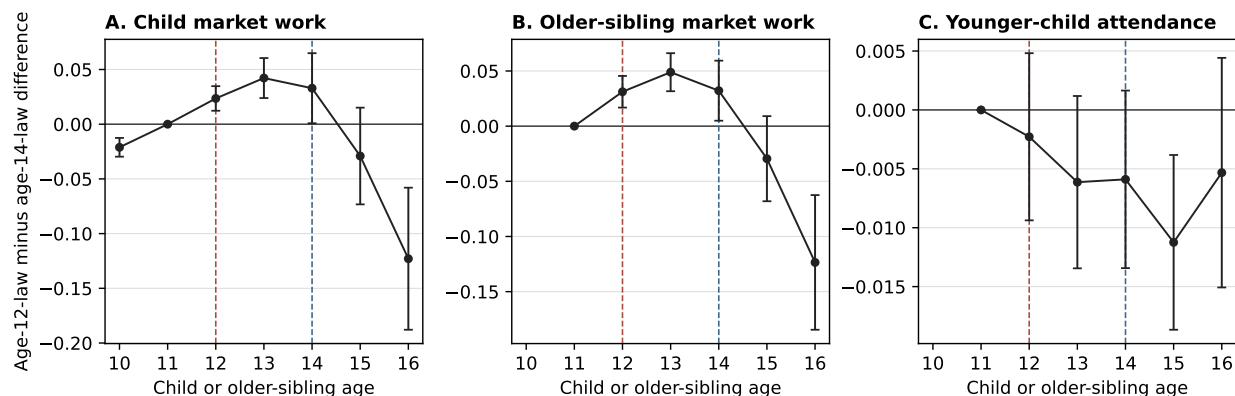
Notes: The treatment is the share of strictly older siblings doing market work. The instruments are older siblings’ mean legal-work exposure interacted with the seven legal regimes. Standard errors are clustered by state. The final row of estimates absorbs state-by-focal-age and exact older-sibling-roster fixed effects. Law-specific first stages appear in Appendix Table 10. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Appendix Table 9 reports the inference and legal-regime robustness checks. The state-clustered standard error is 0.018, giving a 95 percent confidence interval from −0.019 to 0.053. This interval includes zero but not the 0.0655 household benchmark reported by [Manacorda \(2006\)](#). A null-imposed wild-cluster Anderson–Rubin test also does not reject zero ($p = 0.759$) but rejects the household benchmark ($p = 0.006$). The leave-one-state-out estimates range from 0.006 to 0.023. These results do not establish that the sibling effect is exactly zero, but they do reject the published effect size.

The law-specific results clarify what identifies the pooled estimate. The age-12 regime receives

61 percent of the two-stage least-squares weight and the age-14 regime receives 20 percent. Five of the seven attendance reduced forms are negative, and none is significantly positive. Restricting the sample to the 39 states with minimum work ages of 12 or 14 produces an estimate of 0.012 with a standard error of 0.018. We therefore interpret the result as a summary of heterogeneous effects of different legal regimes.

The result also depends on how we control for older-sibling ages across states. Allowing state-specific linear profiles gives an attendance coefficient of -0.029 (0.018). Quadratic profiles leave much less identifying variation and give 0.232 (0.158). With only the common age-12 and age-14 laws, the quadratic estimate is 0.293 (0.187) and the remaining first stage is weak. These more flexible specifications cannot rule out a substantial schooling benefit. Appendix Section D.1 reports the identifying variation and inference in each case.



Coefficients are relative to the age-11 difference. Dashed lines mark the two legal thresholds.

Figure 1: Work and attendance profiles under the age-12 and age-14 laws

Notes: The figure compares states with minimum work ages of 12 and 14. Each point is the age-12-law minus age-14-law difference relative to the difference at age 11; bars are 95 percent confidence intervals based on state-clustered standard errors. Panel A uses all children and plots their own market work by age. Panels B and C retain younger children with exactly one older sibling and plot the older sibling's work and the younger child's attendance by the older sibling's age. Dashed lines mark the two legal thresholds.

Figure 1 makes the comparison of effects across legal regimes visible. Relative to states with a work age of 14, market work rises among children in age-12 states when they reach ages 12 and 13. The same threshold pattern appears for older siblings. Younger children's attendance does not rise when this additional older-sibling work appears; its relative changes are negative at each age from 12 through 16 and statistically different from zero only at age 15. The figure therefore shows a first stage at the relevant threshold without the positive attendance response implied by a sibling financing model.

The first two rows of Table 1 show why it is important to hold older-sibling ages fixed. Specifications with state fixed effects or state-by-focal-age fixed effects, but without the exact older-sibling roster, both produce an attendance coefficient near -0.025 . These estimates partly compare younger children whose older siblings are at different stages of the work-school transition. Once those age differences are removed, the negative relationship disappears.

The detailed time-use categories also provide little evidence that older-sibling work protected younger children’s attendance. The coefficient is -0.009 for work without school, 0.022 when combining work and school, and -0.005 for school without work. The only discernible change is a modest increase in combining work and school, rather than a shift toward school attendance associated with an older sibling’s work.

As a placebo, we reverse the direction and use younger-sibling work to explain the outcomes of older children. The estimates are extremely sensitive to age composition. The attendance coefficient falls from an implausible -1.231 with state fixed effects to -0.168 with a standard error of 0.117 after the exact younger-sibling age roster is held fixed. The timing is also poorly aligned with the mechanism, since an older child’s schooling may already be determined when a younger sibling becomes eligible to work. We report these estimates in the appendix because they show how easily sibling-age differences can be mistaken for a response to sibling work.

The finding is not confined to the original family restrictions. Relaxing the parental-age and family-composition rules expands the directional sample to 4.88 million children and gives an attendance coefficient of -0.013 (0.015). Following linked sons to 1940 also reveals no detectable increase in completed education, although those intervals remain wide. Appendix Sections D.2 and D.3 report these extensions. The next section explains why these findings differ from the large positive effect obtained from a household-average treatment.

6 Why the Household Estimate Differs

The preferred design measures work and legal exposure only among older siblings. The household specification in [Manacorda \(2006\)](#) instead averages across all children aged 10–16. Let $k = 1, \dots, N_f$ index these children in family f . The family work share is

$$\overline{W}_f = \frac{1}{N_f} \sum_{k=1}^{N_f} W_{kf}. \quad (5)$$

Manacorda instruments this share with the average legal exposure of the same children, interacted with indicators for the seven state legal-age regimes. The regression also controls for the focal child’s own legal exposure within each regime, together with the published covariates and state fixed effects. Both the treatment and the instruments therefore include the focal child.

Panel A of Table 2 shows that this household estimate is highly reproducible. The published attendance coefficient is 0.131 . Applying the specification to the deposited microdata gives 0.133 , with the same standard error after rounding, while the full-count estimate is 0.134 . The difference between the household and older-sibling results is therefore not due to sampling error or data construction.

In a family with two children aged 10–16, the full-count coefficient would imply that moving one child into work raised the other child’s attendance by 6.7 percentage points. The household coefficient may not actually reflect that channel of influence, however, because the family work share

and its instruments include the focal child. The issue is clearest when a family contributes only one child aged 10–16, in which case the household measures capture only that child’s own work and legal exposure. The controls for the focal child’s own exposure need not remove this direct response because their coefficients are constrained to be the same across family sizes.

Figure 2 decomposes the full-count estimate by the number of children aged 10–16. The top panel reports each group’s share of the residualized first-stage covariance, while the lower panels show its contribution to the work and attendance coefficients.

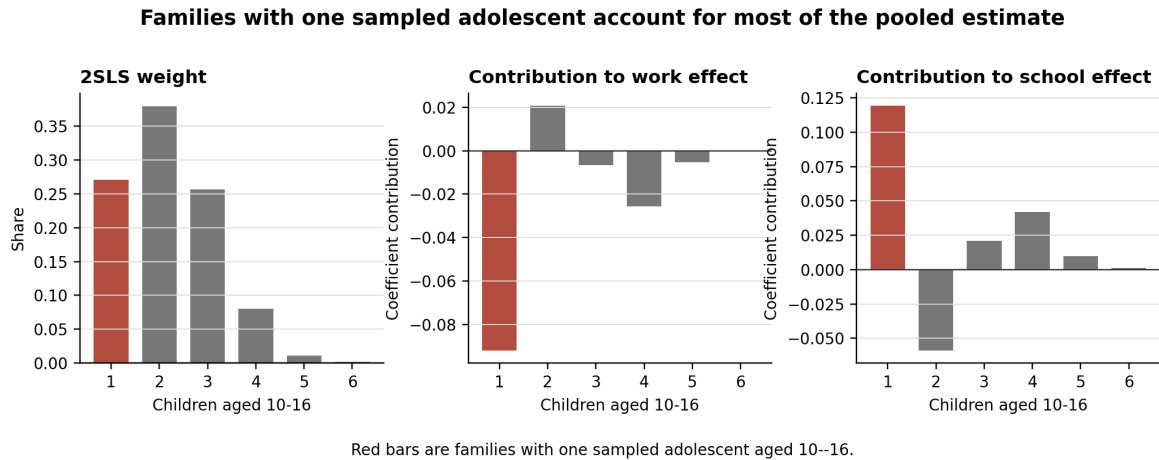


Figure 2: Family-size contributions to the household-average estimate

Notes: The figure decomposes the publication-matched full-count 2SLS estimate by the number of sampled adolescents aged 10–16 in the family. The treatment, instruments, and controls are residualized before constructing the group contributions described in Appendix B. Implicit weights sum to one but need not be positive; the accompanying table reports any negative weights. Families contributing one sampled adolescent are shown in red.

Families with one sampled adolescent account for 1.40 million of the 4.43 million observations and receive 27.1 percent of the two-stage least-squares weight. However, despite this modest weight, they generate 89 percent of the positive attendance coefficient. Again, a family with only one sampled adolescent cannot identify whether one adolescent’s work protected another adolescent’s schooling. Moreover, families contributing two adolescents produce an effect in the opposite (“wrong”) direction.

Panel B of Table 2 applies two corrections. First, it allows the focal child’s response to legal exposure to differ with the number of sampled adolescents. The attendance coefficient changes from 0.134 to -0.084 . Simply excluding families with one sampled adolescent produces an estimate of -0.022 (0.019), as reported in Appendix Table 5. Second, the table removes the focal child from both the work measure and the instruments. The pooled leave-one-out estimate is negative because it compares siblings at different points in the work-school transition. Once the complete sibling-age roster is held fixed, the attendance coefficient is 0.015 (0.026). Among families with two sampled adolescents, it is 0.012 (0.025).

The household coefficient is therefore a reproducible joint response of family work and schooling to child-labor laws, but it does not isolate a transfer across siblings. Its positive attendance effect is generated primarily by families in which the focal child also constitutes the household treatment.

Table 2: From the Household Benchmark to Sibling Comparisons

Specification	Any work	In school	Neither
<i>Panel A. Household benchmark across samples</i>			
Published Table 6	−0.164** (0.075)	0.131* (0.076)	−0.025 (0.064)
Deposited microdata, published inference	−0.166** (0.075)	0.133* (0.076)	−0.026 (0.064)
Full count, state clusters	−0.109*** (0.030)	0.134*** (0.026)	−0.025* (0.014)
<i>Panel B. Separating the focal child from the treatment</i>			
Own-law slopes by family size	0.058*** (0.020)	−0.084*** (0.016)	0.035*** (0.006)
Leave-one-out siblings, pooled	0.090*** (0.008)	−0.143*** (0.011)	0.055*** (0.008)
Leave-one-out, exact sibling-age roster	0.020 (0.013)	0.015 (0.026)	−0.014 (0.016)
Two sampled adolescents, exact sibling age	0.048** (0.022)	0.012 (0.025)	−0.015 (0.017)
Observations, household average	4,428,417		
Observations, exact leave-one-out	3,029,079		

Notes: Panel A reports two-stage least-squares coefficients on the family market-work share. The published and deposited-microdata rows use the article’s inference; the full-count row clusters by state. Panel B first allows the focal child’s legal-exposure response to vary with the number of sampled adolescents and then removes the focal child from the treatment and instruments. Exact-roster models absorb state-by-focal-age and complete sibling-roster fixed effects. Panel B standard errors are clustered by state. The household sample contains 4,428,417 observations; the exact leave-one-out sample contains 3,029,079. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Once the focal child is removed and sibling ages are compared on equal terms, the positive effect disappears.

7 Did Vulnerable Families Benefit?

A small, or null, average effect could conceal a meaningful benefit among families that depended most heavily on child earnings. We cannot identify these families using income-related measures from 1920 because those measures may themselves respond to child work and schooling. Instead, we use family characteristics observed a decade earlier.

Our primary measure ranks the father’s 1910 occupational-income score relative to other linked fathers in the same state. Families below the state median are classified as having low paternal occupational standing. We also distinguish families in which at least one parent was illiterate in 1910. Neither measure is a complete description of family resources, but both predate the work and schooling outcomes by ten years and cannot have been caused by the 1920 treatment.

Table 3 first shows that the main older-to-younger estimate is similar in the linked sample. The attendance coefficient is 0.008 with a standard error of 0.018, which is close to the full-count result. Splitting the sample by the father’s earlier occupational standing produces little difference: the estimate is 0.012 for lower-standing families and 0.007 for higher-standing families, with a standard error of 0.024 in both groups. And again, legal exposure predicts older-sibling work in both groups.

Table 3: Family Background in 1910

1910 family background	Attendance effect	Attendance mean
All linked families	0.008 (0.018)	0.937
Low paternal occupation score	0.012 (0.024)	0.927
Higher paternal occupation score	0.007 (0.024)	0.952
At least one illiterate parent	0.001 (0.048)	0.853
No illiterate parent	−0.001 (0.016)	0.946
Linked observations	1,217,941	
Linked families	846,145	

Notes: Families are linked to their 1910 census characteristics. Low paternal occupation score means below the state median among linked families. Parental illiteracy is measured before the 1920 treatment and outcome. Each row uses the exact older-sibling-age-roster design from Table 1. Standard errors are clustered by state and focal-child age. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

The predetermined parental-literacy split gives the same result. The coefficient on school attendance is 0.001 when at least one parent was illiterate in 1910 and −0.001 when neither parent

was classified as illiterate. Legal exposure predicts older-sibling work in both samples. Restricting the analysis to the 39 states with minimum work ages of 12 or 14 also leaves the estimates near zero in both occupational groups. Appendix Table 11 reports the descriptive full-count splits. In the 1920 literacy split, the coefficient is negative among families with an illiterate parent: -0.094 (0.019), compared with -0.004 (0.047) among other families. Because literacy in 1920 may reflect the same household circumstances that shaped children’s work and attendance, we treat this as descriptive heterogeneity rather than an ideal test. It nevertheless indicates that there is little to no evidence of a positive schooling response among low-resource families.

Nor do the work outcomes suggest that older-sibling work protected younger children along a different attendance margin. Among lower-score families, the coefficients for younger-child work and idleness are both close to zero. Among higher-score families, any work rises by 0.044, largely because more children combine work and school, while idleness remains unchanged. In any case, this pattern does not resemble a school attendance benefit concentrated among families with the fewest resources.

Appendix Table 15 re-weights the linked samples to reproduce the observable composition of the eligible full-count sample. The school attendance estimate becomes 0.024 (0.028) among families with low paternal occupational standing, compared with 0.003 (0.026) among higher-standing families. It becomes 0.016 (0.050) among families with an illiterate parent, compared with -0.002 (0.016) among other families. The first-stage statistics remain above 148 in every re-weighted sample. Re-weighting therefore does not reveal a school attendance benefit among observably vulnerable families, though it is important to note that, because the census linkage is selective, these estimates should not be read as population effects. In other words, re-weighting addresses observed differences between linked and unlinked families, but not differences in unobserved mobility or other types of family disruption.

One possibility is that the broad sibling measure averages over children who are too far in age from the relevant schooling and work margins. Appendix Table 16 narrows the comparison to children aged 10–11 with an older brother aged 12–14. In the full count, the attendance coefficient is -0.064 with a standard error of 0.094. The estimate is -0.180 (0.165) among children with an illiterate parent. It is -0.330 (0.147) among families with low paternal occupational standing, but the corresponding estimate for higher-standing families is -0.076 (0.205), and the difference (-0.255 (0.217)) is not statistically significant. Since legal exposure generates limited identifying variation in these smaller samples, we treat the estimates as suggestive.

8 Conclusion

This paper revisits the claim that work by one child helped finance a sibling’s schooling. We use state minimum-working-age laws and the full-count 1920 Census to separate children’s own responses to the law from effects operating across siblings. The design compares younger children with identical older-sibling age structures across states where the laws induced different amounts of older-sibling

work.

The preferred estimate is 1.7 percentage points, with a state-clustered confidence interval from -1.9 to 5.3 points. The result remains small among families that appeared economically vulnerable in 1910. The linked 1940 estimates also show no detectable increase in completed education, but permit moderately sized gains. More flexible state-specific age profiles weaken identification and cannot rule out substantial attendance benefits. We therefore distinguish the absence of a clear positive sibling effect in the preferred design from a general conclusion that children’s earnings never financed schooling.

We contrast our findings with existing household-level estimates from [Manacorda \(2006\)](#) that imply a sizable intra-household transfer of resources. The large positive attendance coefficient appears in both the original paper’s one-percent sample and our full count, but it is generated largely by families with only one sampled adolescent. Once work is measured among other children and sibling ages are held fixed, the apparent benefit disappears.

These findings concern a specific source of schooling finance, rather than the importance of household resources generally. Unlike a cash transfer, an older child’s employment changes the family’s time allocation as well as its income. Additional earnings could be offset by lost caregiving or other work within the household. Related evidence shows why the form of the intervention matters. In a randomized schooling-transfer experiment in Colombia, untreated siblings attended less when another child received the subsidy ([Barrera-Osorio et al., 2011](#)). The subsidy changed both household resources and the relative incentives to educate different children. Our census data record employment but not the resulting income gain, so we cannot separate an income effect from other household adjustments. The absence of a clear attendance benefit should therefore not be read as evidence that families did not share resources.

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A Benchmark Reconstruction and Sample Details

The deposited archive contains census microdata, law files, and estimation programs. The archived program does not implement the final published specification. It restricts the sample to urban households, uses different parent-age rules, defines work from contemporaneous rather than harmonized occupation codes, and uses one binary legal-eligibility instrument. A model with one instrument cannot produce the six-degree-of-freedom overidentification tests reported in the article.

We reconstruct the published design from the deposited raw extract. We apply the sample rules in the article, use the harmonized occupation definition, construct years at or above the legal work age, and interact family mean exposure with all seven law regimes. We also include the reported controls and clustering. This procedure recovers the published coefficients and standard errors almost exactly and provides the specification used in the full count.

Table 4: Sample and Code Crosswalk

Sample or implementation	Children	Families	Main distinction
Published article	44,402	26,612	Counts reported in Table 6
Archived <code>reg.do</code> sample	54,391	34,118	Urban-only sample, binary eligibility instrument, and earlier variable definitions
Deposited cleaned data	42,478	25,780	Published restrictions, after the archived cleaning program removes households with same-age children
Raw deposited extract	44,451	26,645	Our reconstruction of the published restrictions and seven-instrument design
Deterministic one-percent full count	44,448	26,635	Same construction as the full-count sample
Full-count census	4,428,417	2,652,969	Publication-matched family restrictions applied to the complete census

Notes: The article does not provide row identifiers for its estimation sample, so the 49 children and 33 families separating our raw-extract reconstruction from the published count cannot be listed individually. The close coefficient match in Table 2 shows that these records do not explain the published result. The official article abstract and the archived program describe an urban sample, while the published sample counts are recovered by treating urban status as a control. We therefore report an urban-only sibling estimate separately.

Table 5 reports two additional reconstruction checks. Excluding families with only one sampled adolescent eliminates the positive household estimate. Restricting the sample to urban households instead strengthens the household coefficient but leaves the older-sibling estimate near zero.

Table 5: Household Reconstruction and Urban-Sample Checks

Specification	Attendance effect	Observations
Household model, all families	0.134*** (0.026)	4,428,417
Household model, at least two sampled adolescents	−0.022 (0.019)	3,029,079
Household model, urban households	0.195*** (0.017)	1,964,764
Older-sibling model, urban households	−0.017 (0.015)	711,873

Notes: Standard errors are clustered by state. Household models include the focal child in family work and legal exposure; the second row excludes families contributing only one adolescent aged 10–16. The final row uses only older siblings in the treatment and instruments.

Table 6: Full-Count Sample Composition

Sampled adolescents aged 10–16	Child observations	Families	Share of children
1	1,399,338	1,399,338	0.316
2	1,660,800	830,400	0.375
3	1,006,572	335,524	0.227
4	311,812	77,953	0.070
5	43,920	8,784	0.010
6 or more	5,975	970	0.001
Total	4,428,417	2,652,969	1.000

Notes: Publication-matched full-count sample. Families are two-parent nuclear households satisfying the parent-age and child-age restrictions described in Section 3.

B Formal Two-Stage Least-Squares Decomposition

The household treatment can be written as $\bar{W}_f = N_f^{-1}W_{if} + (N_f - 1)N_f^{-1}\bar{W}_{-i,f}$, where W_{if} is the focal child’s work and $\bar{W}_{-i,f}$ is average work among the other children. To learn which family structures generate the two-stage least-squares estimate, we first partial out the controls and fixed effects. Let M_X be the residual-maker for those controls. Define $\tilde{Y} = M_X Y$, $\tilde{D} = M_X D$, and $\tilde{Z} = M_X Z$, where D is family work and Z contains the seven law-specific family-exposure instruments. The fitted treatment is

$$\hat{D} = P_{\tilde{Z}}\tilde{D} = \tilde{Z}(\tilde{Z}'\tilde{Z})^{-1}\tilde{Z}'\tilde{D}. \quad (6)$$

With one endogenous treatment, the pooled two-stage least-squares coefficient can then be written

$$\hat{\beta} = \frac{\hat{D}'\tilde{Y}}{\hat{D}'\tilde{D}}. \quad (7)$$

Partition the observations into groups g , such as the number of sampled adolescents in the family or the state’s legal-age regime. Define

$$\omega_g = \frac{\hat{D}'_g\tilde{D}_g}{\hat{D}'\tilde{D}}, \quad \hat{\beta}_g = \frac{\hat{D}'_g\tilde{Y}_g}{\hat{D}'_g\tilde{D}_g}. \quad (8)$$

It follows directly that

$$\hat{\beta} = \sum_g \omega_g \hat{\beta}_g. \quad (9)$$

The weights sum to one, but they need not be positive: after residualization, a group can contribute a negative first-stage covariance. Our decomposition therefore reports each group’s numerator, denominator, implicit weight, coefficient, and contribution to the pooled estimate, and explicitly flags negative weights. This is a finite-sample decomposition of the fitted multi-instrument model; it does not treat each group coefficient as a separate causal effect.

Table 7: State-Level and Weak-Instrument-Robust Diagnostics

Diagnostic	Null or estimate	p -value
First-stage partial R^2 , preferred model	0.00269	–
State-clustered Anderson–Rubin, $\beta = 0.0000$	64.789	0.000
State-clustered Anderson–Rubin, $\beta = 0.0655$	211.632	0.000
Wild-cluster Anderson–Rubin, $\beta = 0.0000$	66.168	0.759
Wild-cluster Anderson–Rubin, $\beta = 0.0655$	216.136	0.006
Leave-one-state-out range	[0.006, 0.023]	–
Permutation: older-sibling work	-0.000	0.953
Permutation: younger-child attendance	-0.001	0.383

Notes: Anderson–Rubin tests use the law-specific instruments jointly. The partial R^2 is from the preferred first stage after absorbing controls and fixed effects. The wild-cluster test uses null-imposed Rademacher weights at the state level. The state-law permutation reassigns the observed work-age regimes across states while preserving their frequency and is reported as a design diagnostic rather than literal randomization inference. The benchmark $\beta = 0.0655$ is the attendance effect implied by one additional working child in a two-child family.

The asymptotic and wild-cluster Anderson–Rubin tests disagree about a zero effect. The asymptotic test rejects zero, while the null-imposed wild-cluster test does not. With 48 policy clusters and reduced forms that differ across legal regimes, we place greater weight on the wild-cluster result. The empty confidence set produced by inverting the asymptotic test is therefore evidence that the constant-effect model fits the law-specific moments poorly, not evidence of unusually high precision.

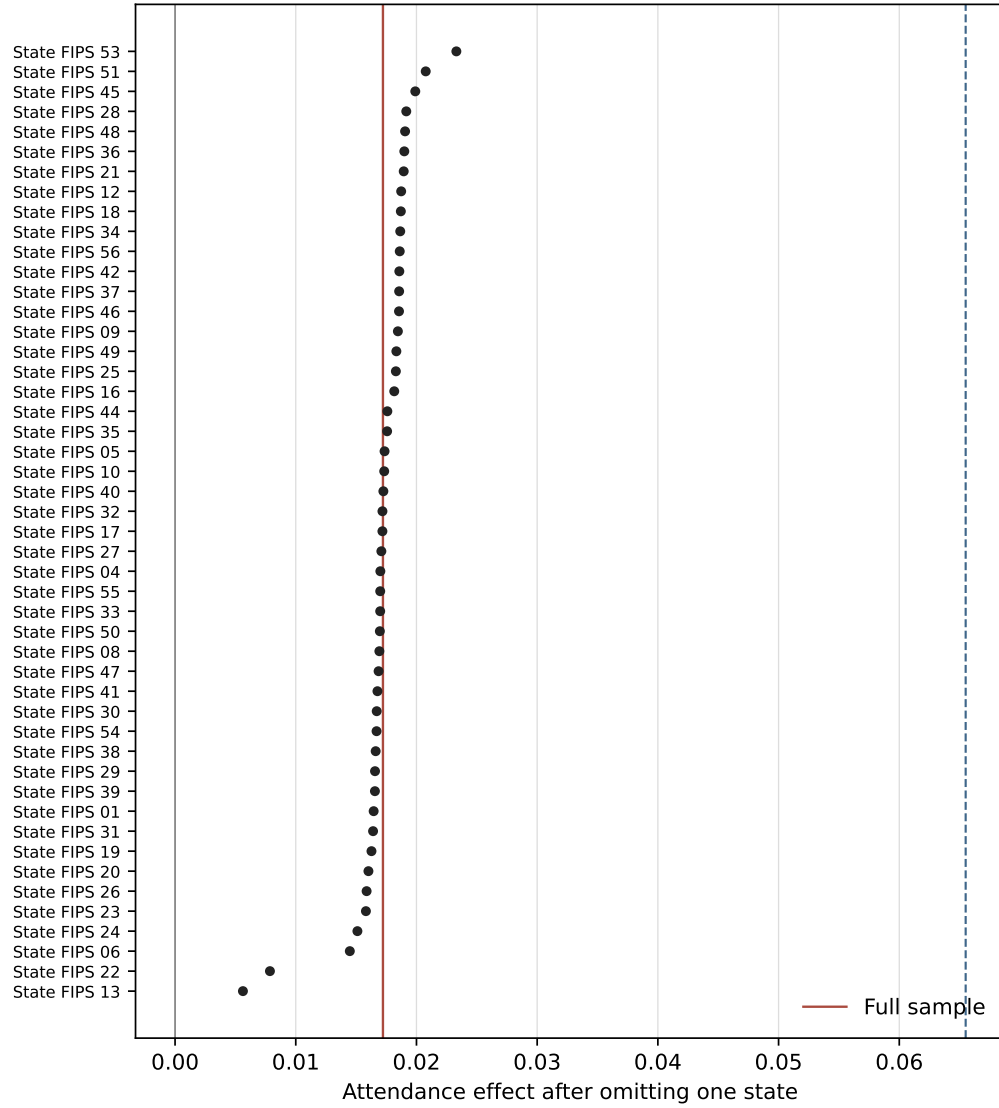


Figure 3: Sensitivity of the attendance estimate to omitting one state

Notes: Each point re-estimates the preferred coefficient after deleting one state's contribution to the residualized sample moments. The solid vertical line is the full-sample estimate; the dashed line is the 0.0655 two-child household benchmark.

Table 8: State Child-Labor and Compulsory-Schooling Laws in 1920

State	Work age	Entry	Leaving	School exempt.	Work permit	Cont. school	Cont. age
Alabama	12	8	16	8	—	No	—
Arizona	14	8	16	8	5	No	—
Arkansas	14	7	15	8	4	No	—
California	14	8	15	8	8	Yes	16
Colorado	14	8	16	8	—	Yes	16
Connecticut	14	7	16	—	5	Yes	16
Delaware	14	7	14	—	—	No	—
Florida	8	8	14	—	—	No	—
Georgia	12	8	14	4	—	No	—
Idaho	15	7	16	8	8	No	—
Illinois	14	7	16	—	5	Yes	16
Indiana	14	7	16	—	5	Yes	16
Iowa	14	7	16	8	6	No	—
Kansas	14	8	16	8	8	No	—
Kentucky	14	7	16	—	5	No	—
Louisiana	7	7	14	8	—	No	—
Maine	14	7	15	—	5	No	—
Maryland	13	7	15	—	5	No	—
Massachusetts	14	7	16	7	4	Yes	—
Michigan	14	7	16	8	6	No	—
Minnesota	14	8	16	8	—	No	—
Mississippi	12	7	14	8	—	No	—
Missouri	14	8	16	—	—	No	—
Montana	14	8	16	—	—	No	—
Nebraska	14	7	16	8	8	Yes	16
Nevada	14	8	16	8	—	No	—
New Hampshire	14	8	16	8	5	Yes	—
New Jersey	14	7	16	—	5	No	—
New Mexico	—	7	14	—	—	No	—
New York	14	8	16	8	8	Yes	16
North Carolina	12	8	14	—	—	No	—
North Dakota	14	7	15	8	—	No	—
Ohio	15	8	16	6	6	Yes	16
Oklahoma	14	8	16	—	—	No	—
Oregon	14	9	15	8	—	No	—
Pennsylvania	14	8	16	—	6	Yes	16
Rhode Island	14	8	16	8	—	No	—
South Carolina	14	8	16	—	—	No	—
South Dakota	15	8	16	8	—	No	—
Tennessee	14	8	16	8	—	No	—
Texas	12	8	14	—	4	No	—
Utah	12	7	16	8	—	No	—
Vermont	15	8	16	8	6	No	—
Virginia	14	8	12	4	—	No	—
Washington	12	8	16	8	8	No	—
West Virginia	14	8	15	4	4	No	—
Wisconsin	14	7	16	8	5	Yes	17
Wyoming	7	7	14	—	—	No	—

Notes: Work age is the minimum legal employment age. Entry and leaving are the compulsory-schooling ages; school exemption and work permit report completed-schooling requirements. A dash denotes no recorded requirement. Source: deposited state-law files and exceptional state codings described in Section 2.

C Additional Sibling Results

Appendix Table 9 reports alternative inference and legal-regime specifications, Appendix Table 10 reports the law-specific first stages and reduced forms, and Appendix Table 11 reports descriptive full-count heterogeneity.

Table 9: State-Level Inference and Legal-Regime Robustness

Specification	Attendance effect	Mean	Observations
Exact older-sibling roster	0.017 (0.018)	0.927	1,756,540
State-specific sibling-age profiles	0.232 (0.158)	0.927	1,756,540
Seven legal regimes	0.017 (0.018)	0.927	1,756,540
Exclude ages 0, 7, and 8	0.011 (0.017)	0.931	1,692,523
Work ages 12 and 14, two instruments	0.012 (0.018)	0.928	1,562,713
Work ages 12 and 14, pooled exposure	0.603 (0.843)	0.928	1,562,713
Urban households	−0.017 (0.015)	0.953	711,873

Notes: Each entry is a two-stage least-squares coefficient on the older-sibling market-work share. Standard errors in parentheses are clustered by state. The first row includes state-by-focal-age and exact older-sibling-roster fixed effects. The state-profile model additionally permits state-specific quadratic profiles in mean older-sibling age. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

We report the reverse younger-to-older models only as a diagnostic. With state fixed effects, the coefficients are 0.807 (0.122) for work, −1.231 (0.140) for school, and 0.203 (0.060) for idleness. State-by-focal-age fixed effects reduce them to 0.119 (0.047), −0.201 (0.055), and 0.004 (0.026). Exact younger-sibling age-roster fixed effects give 0.030 (0.092), −0.168 (0.117), and −0.003 (0.068). The sample contains 1,752,840 observations in 1,242,863 families. The instability shows why sibling-age composition matters and why the younger-to-older timing is poorly suited to a financing interpretation.

Figure 4 traces the progression from household averages to exact-roster sibling comparisons in both the deposited sample and the full count. The one-percent sample is too small to estimate the sibling specifications precisely, but the estimates move in the same direction as the full-count results.

The broader leave-one-out specifications produce the same small attendance effect once sibling ages are held fixed. Reassigning the common age-12 and age-14 laws across states produces permutation p -values of 0.953 for older-sibling work and 0.383 for younger-child attendance. Restricting the linked analysis to the 39 states with those two age limits also leaves the estimates near zero in both paternal-occupation groups. Reduced forms differ across legal regimes in several pooled

Table 10: First Stages, Reduced Forms, and Weights by Legal Regime

Minimum work age	States	First stage	Attendance reduced form	2SLS weight
Unrecorded	1	0.007 (0.006)	0.000 (0.001)	2.3%
7	2	0.027*** (0.006)	−0.005*** (0.001)	4.5%
8	1	0.032*** (0.006)	−0.001* (0.001)	1.4%
12	7	0.022*** (0.006)	−0.003*** (0.001)	61.0%
13	1	0.092*** (0.007)	0.001 (0.001)	4.8%
14	32	0.076*** (0.014)	−0.002** (0.001)	20.0%
15	4	0.082*** (0.016)	−0.000 (0.001)	6.0%

Notes: The first-stage and reduced-form columns report coefficients on mean older-sibling legal exposure within each minimum-work-age regime. Standard errors are clustered by state. Weights are computed from the residualized multi-instrument first stage and sum to one. A dagger marks a negative weight. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 11: Full-Count Heterogeneity in the Older-Sibling Estimate

Comparison	Group 0	Group 1	Difference
Rural / Urban	−0.126** (0.050)	−0.059** (0.028)	0.068*** (0.026)
Nonfarm / Farm	−0.021 (0.031)	−0.046 (0.067)	−0.025 (0.040)
Boys / Girls	0.007 (0.017)	0.023 (0.020)	0.016 (0.010)
Literate parents / At least one illiterate parent	−0.004 (0.047)	−0.094*** (0.019)	−0.090** (0.037)

Notes: Joint interacted two-stage least-squares models estimate both group coefficients and their difference in a common sample. Standard errors are clustered by state. Parental illiteracy is observed in 1920; the predetermined 1910 measures are reported separately. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

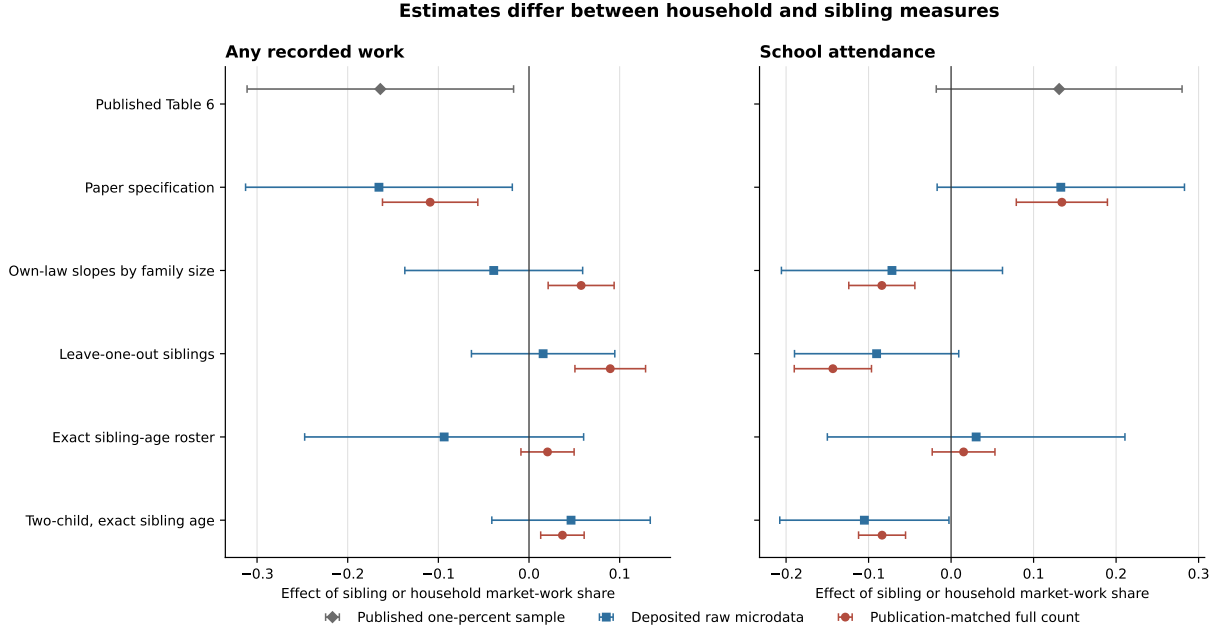


Figure 4: Household-average and sibling specifications

Notes: Points report 2SLS coefficients and 95 percent confidence intervals. Blue squares use the deposited one-percent census microdata; red circles use the publication-matched full count. The published Table 6 coefficient is shown in gray. Exact-roster models hold the complete set of sibling ages fixed.

specifications, so we interpret the coefficients as weighted summaries of the state laws in place in 1920 rather than as a constant response across all regimes.

D Age Profiles, Family Scope, and Later Schooling

D.1 State-specific age profiles

The exact-roster design holds older-sibling ages fixed across states, but does not absorb state-specific differences in how those ages predict younger-child attendance. We examine this remaining restriction by adding state-specific linear and quadratic terms in mean older-sibling age. Because focal children are at least 10, their strictly older siblings are at least 11. Legal exposure under thresholds 0, 7, 8, and 12 is therefore linear in older-sibling age throughout this sample. The state-specific linear controls absorb those four instruments. We remove absorbed columns before normalizing the residual instruments, so numerical rounding does not create spurious identifying variation.

The linear specification retains 37.5 percent of the baseline fitted-work variance. Adding quadratic terms reduces this to 3.0 percent. Its attendance coefficient is 0.232, with a state-clustered 95 percent interval from -0.079 to 0.542 . Restricting the quadratic model to states with the common thresholds leaves only the age-14 instrument, with a first-stage Wald statistic of 7.6. The coefficient is 0.293, and the interval runs from -0.073 to 0.660 . A wild-cluster Anderson–Rubin test rejects neither zero ($p = 0.178$) nor the 0.0655 benchmark ($p = 0.250$) in this common-law specification.

Table 12: Identification and Age Profiles

Outcome	Exact roster	Linear profiles	Quadratic profiles	Quadratic ages 12 and 14
School attendance	0.017 (0.018)	−0.029 (0.018)	0.232 (0.158)	0.293 (0.187)
Observations	1,756,540	1,756,540	1,756,540	1,562,713
Families	1,242,863	1,242,863	1,242,863	1,104,895
States	48	48	48	39
Excluded instruments	7	3	3	1
First-stage Wald χ^2	104,891.3	235.4	34.6	7.6
Fitted-work variance (%)	100.0	37.5	3.0	2.1

Notes: The treatment is the share of strictly older siblings aged 10–16 in market work. All models retain state-by-focal-age and exact older-sibling-age-roster fixed effects. Linear and quadratic profiles are polynomials in mean older-sibling age interacted with state. Algebraically absorbed instruments are excluded before normalization. Fitted-work variance is expressed relative to the first column. The Wald statistic is chi-squared, not an F statistic. Standard errors clustered by 1920 state are in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

The preferred estimate is therefore not evidence against a large benefit under every admissible age profile.

The multiple-instrument quadratic model rejects both zero and the household benchmark in joint Anderson–Rubin tests, despite the wide interval around its pooled coefficient. These tests assess a common coefficient across law-specific moments. The disagreement with the pooled interval cautions against interpreting their rejection as evidence of a precisely estimated positive or negative sibling effect. Residualizing the instruments also means that a state’s contribution to the pooled estimate need not vanish when its own raw instrument is absorbed.

D.2 Broader family samples

We first relax the original parental-age limits, allowing both parents to be aged 18–99 while retaining the published family-composition rule. We then admit adult children and extended-family structures. Siblings continue to share the same identified mother and father, and the treatment remains the market-work share among strictly older siblings aged 10–16. The original restrictions concern the census family unit, so some other household members were already present. All three comparisons below control for the number of children over 16 and other household members.

The expanded panel reproduces all 4,428,417 children in the original sample without any change in their work, attendance, family identifiers, or birth order. The directional sample increases from 1,756,540 to 2,274,394 children when parental ages are relaxed and to 4,876,679 when family composition is also relaxed. The attendance coefficients are 0.017 (0.018), 0.008 (0.020), and −0.013 (0.015), respectively. The broadest sample’s 95 percent interval runs from −0.043 to 0.016. A wild-cluster Anderson–Rubin test does not reject zero ($p = 0.259$) but rejects a five-percentage-point

Table 13: School Attendance Under Broader Family Restrictions

Outcome	Original	Parental ages 18–99	Expanded families
School attendance	0.017 (0.018)	0.008 (0.020)	−0.013 (0.015)
Observations	1,756,540	2,274,394	4,876,679
Families	1,242,863	1,603,295	3,341,074
States	48	48	48

Notes: The treatment is the share of strictly older siblings aged 10–16 in market work. All models retain state-by-focal-age and exact older-sibling-age-roster fixed effects. The samples are nested and require two identified co-resident parents. All three models also control for children over 16 and other household members. The second row relaxes parental age restrictions; the third also relaxes family composition. Standard errors clustered by 1920 state are in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

benefit ($p < 0.001$). Broader family coverage therefore does not reveal the proposed attendance benefit under the preferred age controls. This exercise does not cover one-parent families or remove the identifying restrictions on state-specific age patterns.

D.3 Completed education in 1940

The eligible 1920 sample contains 891,197 younger boys and 865,343 younger girls. Valid 1940 links are available for 51.6 percent of boys and 14.8 percent of girls. The attainment sample contains 451,232 sons with observed education. We measure years of education, capped at 17, and indicators for completing grades 8 and 12. Missing education codes are not treated as noncompletion. Every outcome uses the same 1920 older-sibling treatment, instruments, and fixed effects as the preferred attendance model.

Older-sibling work does not produce a detectable increase in completed schooling. The years-of-education coefficient is -0.257 (0.427), with a 95 percent interval from -1.095 to 0.580 . Reweighting gives -0.286 (0.424), with an interval from -1.117 to 0.544 . Grade 8 and grade 12 estimates are also imprecise. These results do not establish the absence of a long-run benefit. In particular, both the conventional intervals and wild-cluster Anderson–Rubin tests permit a gain of half a year of education.

The reweighting targets eligible 1920 sons and balances state, focal age, the number and mean age of older siblings, younger children, urban and farm status, race, parental literacy and immigration, housing tenure, and parental ages. Capping weights at the 99th percentile reduces the largest difference across these margins from 5.94 to 0.34 percentage points and leaves an effective sample of 410,060 sons. It cannot correct selection on unobserved determinants of survival or linkage. Legal exposure predicts lower male linkage in the low-age regimes, with reduced-form slopes of -0.009 , -0.006 , and -0.005 under thresholds 0, 7, and 8. All three are statistically different from zero at

Table 14: Older-Sibling Work and Completed Education in 1940

Outcome	Linked sons	Reweight
Attendance, 1920	0.007 (0.022)	0.023 (0.030)
Years of education, 1940	−0.257 (0.427)	−0.286 (0.424)
Completed grade 8	−0.052 (0.063)	−0.053 (0.060)
Completed grade 12	0.014 (0.048)	0.018 (0.044)
Observations	451,232	451,232
Families	405,642	405,642
States	48	48

Notes: The treatment is the share of strictly older siblings aged 10–16 in market work. All models retain state-by-focal-age and exact older-sibling-age-roster fixed effects. Focal children are sons with unambiguous 1940 links and observed attainment. Years of education are capped at 17. Grade 8 and grade 12 indicate completion. Reweighting balances observed 1920 characteristics; it does not correct unobserved linkage selection. Standard errors clustered by 1920 state are in parentheses. $*p < 0.10$, $**p < 0.05$, $***p < 0.01$.

the five-percent level. Female linkage is also related to legal exposure in several regimes. The 1920 attendance coefficient in the same linked sample is 0.007 (0.022), or 0.023 (0.030) after reweighting. Law-specific attendance moments remain heterogeneous, so the small pooled coefficient should not be read as evidence that linkage is innocuous.

E Predetermined-Linkage Selection

The linked-background sample contains 2,896,783 child records before the directional and nonmissing-background restrictions. We exclude 3,694 records because children in the same 1920 family map to conflicting 1910 origins. Compared with the unlinked full-count sample, linked children are more likely to come from a farm family (39.1 versus 27.3 percent) and to attend school (90.1 versus 86.2 percent). They are also less likely to have an illiterate father (5.1 versus 13.3 percent).

We construct family-level calibration weights separately for the paternal-occupation and parental-literacy samples. The target is the full-count sample of younger children with at least one strictly older sibling. Within each analysis sample, iterative proportional fitting matches the target distributions of state, sampled-child ages, the number of younger children, sex composition, urban and farm status, race, parental ages, literacy, immigrant status, and housing tenure. Every target category has linked observations. Before weighting, the largest difference across these margins is 4.1 percentage points. After capping weights at the 99th percentile, the largest remaining difference is 0.16 percentage points. The trimmed weights range from approximately 0.57 to 1.95. They retain effective sample sizes of 1.16 million observations in the paternal-occupation sample and 1.20 million in the parental-literacy sample.

Table 15: Reweighting the Linked 1910 Sample

1910 family background	Unweighted	Reweighted
All linked backgrounds	0.008 (0.018)	0.014 (0.020)
Low paternal occupation score	0.012 (0.024)	0.024 (0.028)
Higher paternal occupation score	0.007 (0.024)	0.003 (0.026)
At least one illiterate parent	0.001 (0.048)	0.016 (0.050)
No illiterate parent	−0.001 (0.016)	−0.002 (0.016)
<i>Calibration diagnostics</i>		
Paternal-occupation sample: effective observations	1,159,371 of 1,217,941	
Parental-literacy sample: effective observations	1,200,413 of 1,259,508	

Notes: Entries are two-stage least-squares coefficients for younger-child attendance, with standard errors clustered by state and focal-child age in parentheses. The reweighted estimates calibrate linked families to the eligible full-count 1920 sample on state, sampled-child age composition, the number of younger children, sex composition, urban and farm status, race, parental ages, literacy, immigrant status, and housing tenure. Weights are capped at the 99th percentile. Reweighting addresses selection on these observed characteristics, not selection on unobserved determinants of census linkage. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 15 compares the weighted and unweighted attendance estimates. The coefficients remain small and imprecise in the full linked sample and in both vulnerability splits. The weighting exercise shows that observable differences in the composition of linked families do not explain the result. It cannot correct selection on unobserved determinants of linkage or recover families that had dissolved before 1920. We therefore continue to interpret the linked estimates as a targeted heterogeneity test rather than as a replacement for the full-count result.

F Siblings Near the Legal Work Threshold

The most direct marginal-schooling test would study focal children aged 14 or 15. The exact-roster design cannot identify that comparison under the two main state laws. Every strictly older sibling of a 14- or 15-year-old is already legally exposed in states with minimum work ages of 12 or 14. Once state and the exact older-sibling ages are held fixed, no excluded variation remains. The common-law first stage is also weak for focal children aged 12 because nearly all of their older siblings are exposed in both regimes.

We therefore focus on children aged 10–11 and potential earners aged 12–14. These younger children are below the usual school-leaving margin, but attendance is substantially lower in families with an illiterate parent and in several linked low-resource groups. The excluded instruments interact the older siblings’ mean legal exposure with indicators for the two common minimum work ages. Each model absorbs state by focal-child age and sex and the exact age-by-sex roster of all older siblings.

Table 16: Younger Children and Siblings Near the Legal Work Threshold

Sample and potential earner	In school	Any work	Neither
<i>Panel A. Full-count comparisons</i>			
Older brother, children aged 10–11	−0.064 (0.094)	0.127 (0.132)	0.032 (0.097)
Older sister, children aged 10–11	−0.126 (0.138)	−0.009 (0.099)	0.087 (0.141)
Older brother, children aged 10–12	−0.046 (0.088)	0.135 (0.119)	0.030 (0.089)
<i>Panel B. Family-background comparisons</i>			
At least one illiterate parent	−0.180 (0.165)	−0.046 (0.204)	0.225 (0.152)
Low paternal occupational standing	−0.330** (0.147)	0.132 (0.161)	0.264** (0.133)
Higher paternal occupational standing	−0.076 (0.205)	0.031 (0.130)	0.015 (0.200)
Father in farm work in 1910	−0.343** (0.164)	0.123 (0.146)	0.163 (0.116)
Father outside farm work in 1910	−0.180 (0.175)	−0.028 (0.216)	0.223 (0.172)
Low paternal standing, older sister	−0.009 (0.162)	0.215 (0.137)	−0.073 (0.145)

Notes: The focal children are aged 10–11 unless otherwise noted. Potential earners are strictly older siblings aged 12–14. The table uses states with minimum work ages of 12 or 14 and instruments the sibling’s market work with law-specific mean legal exposure. Every model absorbs state by focal-child age and sex and the exact age-by-sex roster of all older siblings. Low paternal occupational standing is measured in 1910 relative to the state median; parental illiteracy in its row is measured in 1920. Standard errors are clustered by state and focal-child age. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Panel A shows no positive attendance effect in the full count. Panel B selects families with lower attendance or fewer resources. The low-paternal-standing and farm-family estimates are

negative, but neither coefficient is statistically different from its comparison-group estimate. The formal low-minus-high paternal-standing difference is -0.255 (0.217), and the farm-minus-nonfarm difference is -0.163 (0.242).

The low-status attendance estimate is accounted for mainly by children recorded as neither working nor attending school. School without work falls by 0.396 (0.194), while work without school rises by 0.066 (0.060), combining work and school rises by 0.066 (0.164), and neither work nor school rises by 0.264 (0.133). The attendance estimates are nearly identical for younger boys and girls. These patterns do not isolate farm labor, domestic work, or another measured substitution margin, so we do not interpret the negative subgroup coefficient as a separate causal finding.

G Data and Code Availability

The deposited replication archive for [Manacorda \(2006\)](#) contains the original one-percent extract, law files, and archived programs. Our reconstruction code, sample crosswalks, estimation scripts, and nonrestricted output tables will be deposited with the paper. The full-count census and Census Tree links are used under their respective data-use conditions and cannot be redistributed; the repository will instead include the extract definitions, variable crosswalks, and code needed to rebuild the analysis for authorized users.